Software Requirements Specification (SRS) for SportStats Notification Engine

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1. Introduction

1.1 Purpose

This document specifies the requirements for the "SportStats Notification Engine." This tool is designed to provide real-time updates via SMS and email to interested parties (e.g., race directors, parents, or sweep teams) when an athlete crosses a RaceResult decoder.

1.2 Scope

- **Project Identification:** SportStats Notification Engine.
- **Project Goals:** To provide a lightweight tracking solution for smaller races that lack the infrastructure for comprehensive live tracking.
- **Target Users:** Race Directors (tracking first male/female, sweep), Parents/Guardians (child tracking), and Event Staff.
- **Inclusions:** Real-time data ingestion from RaceResult 14, automated notification triggering, and a subscriber mapping database.
- **Exclusions:** This system is not a public results portal or a high-latency statistical analysis tool.

1.3 Definitions and Abbreviations

- **BID:** User bib ID.
- **RaceResult 14:** The timing software that exports bib crossing data.

- **Interested Party:** Any individual registered to receive a notification for a specific BID.

2. Overall Description

2.1 Product Perspective

The Notification Engine acts as a middleware layer between the RaceResult 14 software and external communication APIs (e.g., Twilio for SMS, SendGrid for Email). It mirrors the architectural pattern of previous real-time tools like the Hype Screen but focuses on external communication rather than visual display.

2.2 Product Functions

The system will:

- Ingest live BIB crossing data (BibID) from RaceResult 14 via specified ports.
- Map received BibIDs to a pre-loaded "Subscriber List."
- Determine the notification type (SMS/Email) based on subscriber preference.
- Dispatch automated messages upon crossing detection.
- Log all sent notifications for audit purposes.

2.3 User Characteristics

- **Administrators:** Sportstats staff responsible for setting up the local ingestion port and loading the subscriber CSV.
- **Subscribers:** End-users receiving the SMS/Email alerts.

3. Specific Requirements

3.1 Functional Requirements

3.1.1 Data Integration

- **FR-1.1:** The system SHALL connect to RaceResult 14 via localhost on a specified port to receive real-time bib scan data.

- **FR-1.2:** The system SHALL ingest a CSV mapping of BibIDs to subscriber contact information (SMS numbers and Email addresses).

3.1.2 Notification Logic

- **FR-2.1:** Upon a valid BibID read, the system SHALL trigger notifications to all mapped subscribers for that specific BID within 5 seconds.
- **FR-2.2:** The system SHALL support customizable message templates (e.g., "Athlete [Name] has just crossed the Finish Line!").

3.1.3 Communication Protocols

- **FR-3.1:** The system SHALL integrate with an SMS gateway to deliver text alerts.
- **FR-3.2:** The system SHALL integrate with an SMTP or API-based email service for electronic mail alerts.

3.2 Non-Functional Requirements

ID	Category	Requirement Description
NFR-1	Latency	Notifications SHALL be dispatched within 5 seconds of the decoder read ingestion.
NFR-2	Reliability	The system SHALL maintain a local queue to retry notification delivery in case of momentary API outages.
NFR-3	Scalability	The architecture SHALL support concurrent notification bursts of up to 200 messages per minute.

ID	Category	Requirement Description
NFR-4	Security	Subscriber contact information SHALL be encrypted at rest and not shared with third parties.

4. Technical Constraints

- **Dependencies:** Stable internet connection at the finish line or timing station to reach external communication APIs.
- **Input Format:** RaceResult 14 (RR14) must be configured to export raw BID data to localhost on port 3099.

Approved By: Person
Date: Date